TECHNICAL SPECIFICATION

Electrical Signature Analysis (ESA)

Condition Monitoring System

Tender Requirements & Evaluation Criteria

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Executive Summary

This document specifies the technical requirements for an Electrical Signature Analysis (ESA) condition monitoring system for rotating equipment. It is intended for use in formal tenders, RFPs, and technical evaluations.

The specification covers signal acquisition, fault detection, analytics, service delivery, deployment, integration, and vendor qualification. Requirements are structured as Mandatory (must-have) or Preferred (best-practice targets), enabling fair evaluation across different proven ESA approaches while maintaining a high technical bar.

ESA monitors motor current and voltage signatures at the Motor Control Cabinet (MCC), enabling predictive maintenance of assets that are physically inaccessible to conventional vibration sensors: submerged pumps, equipment in explosive atmospheres (ATEX zones), enclosed or high-temperature machines, and remote installations.

Scope exclusions: This specification targets systems capable of full Electrical Signature Analysis (current + voltage). It explicitly excludes RMS-only power monitors, energy-disaggregation / NILM systems, and portable survey tools. Where current-only Motor Current Signature Analysis (MCSA) is acceptable, this is noted per-requirement.

Interpretation and Equivalence

Where numerical thresholds are stated in this specification, they represent minimum capability expectations based on current industry best practice. Alternative implementations delivering demonstrably equivalent diagnostic capability may be accepted, subject to documented technical justification and field evidence.

This clause preserves evaluator discretion to accept technically sound approaches that satisfy the intent of a requirement through different means. It does not permit omission of mandatory capabilities or substitution without evidence.

Key Requirements Summary

The table below provides a scannable overview for procurement and executive review. Detailed specifications follow in subsequent sections.

Domain	Mandatory Requirement	Preferred Target
Signal Acquisition	≥16 kHz sampling, ≥16-bit effective resolution, 3-phase current + voltage	≥40 kHz sampling, ≥24-bit ADC, configurable acquisition windows
Fault Detection	Coverage of electrical, mechanical, transmission, and load/process faults (electrically observable)	>85% confirmed detection rate across fault categories (measured over 12-month evaluation)
Analytics	Adaptive, trend-based learning diagnostics. No fixed-threshold-only systems.	Demonstrated field validation on ≥1,000 confirmed fault events
Service Delivery	Expert validation of all alerts before customer notification; outputs advisory only	<2% false-positive rate on validated alerts (target <1%)

Domain	Mandatory Requirement	Preferred Target
Monitoring Mode	Permanent online monitoring for critical assets	Continuous streaming with local edge buffering
Deployment	MCC-based installation	<60 min per asset typical installation time
Scalability	Platform supports ≥1,000 assets; multi-site visibility	Demonstrated enterprise deployment at comparable scale
Security	TLS 1.2+ in transit, AES-256 at rest, ISO 27001 or SOC 2 (Security + Availability)	Annual independent penetration testing

Why These Requirements Matter

ESA fault signatures manifest as gradual spectral changes, not abrupt threshold exceedances. A bearing defect, for example, modulates the motor load torque at characteristic frequencies. These modulations appear as sidebands in the current spectrum that grow slowly over weeks or months. Static thresholds set wide enough to avoid false alarms will miss genuine developing faults; set tight enough to catch faults, they fire on normal process variation.

Effective ESA analytics must therefore learn per-asset baselines, track trends over time, and distinguish process-driven load changes from fault-driven spectral shifts. The requirements that follow enforce this principle without prescribing a specific algorithm family — model-based (physics), statistical, and machine-learning approaches can all satisfy the specification if they demonstrate adaptive behaviour and validated field results.

Raw, unvalidated alerts erode maintenance-team trust and cause alarm fatigue. Expert-validated, actionable insights are the difference between a monitoring system that drives value and one that is eventually ignored.

1. Signal Acquisition Requirements

High-quality signal acquisition is foundational. Insufficient sampling rates, resolution, or channel coverage produce unrecoverable information loss upstream of any analytics engine.

Parameter	Mandatory Minimum	Preferred	Optimal / Extended	Rationale / Physics Basis
Sampling Rate	≥16 kHz per channel	≥20 kHz per channel	≥96 kHz per channel	Nyquist: 16 kHz sampling resolves frequencies to 8 kHz, covering the primary motor-fault band (0–5 kHz) and lower-order bearing-related load-modulation harmonics. 20 kHz extends useful bandwidth to 10 kHz. 40 kHz captures high-order harmonics up to ~20 kHz. This tiered structure reflects that many proven ESA systems operate effectively at 16 kHz while higher rates provide additional margin.
ADC Resolution	≥12-bit nominal (ENOB ≥11 bits)	24 bit nominal, ≥18.8 bit ENOB		Theoretical SQNR = 6.02×N + 1.76 dB, giving ~98 dB at 16-bit. In practice, effective resolution is lower due to noise, non-linearity, and thermal drift. Vendors shall state an effective number of bits (ENOB), not only nominal bit depth. 24-bit ADCs typically deliver 18–21 ENOB at industrial temperature ranges.
Measurement Channels	3-phase current AND 3-phase voltage [MANDATORY]	Simultaneous sampling across all channels		Current-only measurement (MCSA) cannot detect supply-side faults, voltage unbalance, power-factor degradation, or perform efficiency analysis. Full ESA requires both. See Appendix B.
Acquisition Window	≥10 seconds minimum per snapshot	≥30 seconds		Spectral resolution Δf = 1/T. A 10 s window yields 0.1 Hz resolution — sufficient for rotor-bar sideband separation at 2×s×f for most 4-pole motors. Longer windows (60) improve resolution to ≤0.017 Hz, valuable for slow-speed machines and close sideband separation.
Frequency Analysis Range	0–5 kHz (analysed)	0–10 kHz or higher		Primary motor faults (rotor bars, eccentricity, stator winding) appear as sidebands in the 0–2.5 kHz band. Some bearing characteristic frequencies and their load-modulation harmonics can extend to 5–10 kHz. Frequencies above 10 kHz may

Parameter	Mandatory Minimum	Preferred	Optimal / Extended	Rationale / Physics Basis
				capture additional high-order bearing-related modulation effects where electrically observable.
Anti-Aliasing Filter	Hardware anti-aliasing filter at or below Nyquist/2	Configurable filter with ≥−60 dB/decade roll-off		Without proper anti-aliasing, high-frequency noise folds into the analysis band, creating phantom spectral peaks that can obscure true fault signatures.

Note on current-only (MCSA) acceptability: *For applications where only mechanical fault detection is required (no supply-quality or efficiency analysis), current-only MCSA may be considered. The vendor shall clearly state which fault categories require voltage channels. Any proposal offering current-only measurement must document the diagnostic limitations this imposes.*

Exclusion Criteria — Signal Acquisition

- ✖ RMS-only or averaged-value systems (discard waveform information essential for spectral analysis).
- ✖ Single-phase measurement only (cannot detect phase imbalance or inter-phase fault signatures).
- ✖ Sampling rate below 12 kHz or unspecified (violates Nyquist for most bearing-related load-modulation harmonics).
- ✖ Effective resolution below 12-bit nominal (ENOB ≥11 bits)
- ✖ No hardware anti-aliasing filter.
- ✖ Energy-disaggregation / NILM approaches that infer machine state from aggregate power consumption.

2. Fault Detection Coverage

A comprehensive ESA system must detect faults across the entire drivetrain: electrical supply, motor, mechanical transmission, and driven load. Narrow coverage creates blind spots that result in undetected failures.

2.1 Required Fault Categories

Fault Category	Specific Faults (Mandatory Coverage)	Detection Mechanism
Electrical Faults	Stator winding faults (turn-to-turn, phase-to-phase, phase-to-ground); rotor bar defects/cracks; voltage unbalance; current unbalance; phase-loss; over-current; harmonic distortion; insulation degradation trends	Negative-sequence and zero-sequence current components analysis, rotor-bar sideband analysis (f ± 2sf), current and voltage harmonic spectrum analysis, current and voltage unbalances
Mechanical Faults	Bearing degradation (inner race, outer race, rolling element, cage); mechanical unbalance; shaft misalignment (angular and parallel); soft foot; rotor eccentricity (static and dynamic)	Bearing-related load-modulation effects and high-order harmonics observable in current and power signatures at bearing characteristic frequencies (BPFO, BPFI, BSF, FTF); eccentricity sidebands at f ± fr; unbalance and misalignment via torque-modulation sidebands and its harmonics.
Transmission Faults	Coupling eccentricity/wear; belt wear/slip/tension; gearbox tooth damage; pulley/sheave issues and misalignments; chain wear.	Torsional oscillation signatures at belt/gear mesh frequencies and their sidebands in the current spectrum.
Load / Process Faults	Pump cavitation; pump blockage; impeller damage; valve restriction; fan blade damage/build-up;	Characteristic load-oscillations frequencies; power-demand anomalies; efficiency degradation trends using current + voltage, abnormal operating time patterns (run-time, start/stop frequency, duty-cycle variations)

2.2 Detection Performance Targets

The following are evaluation targets, not guaranteed absolutes. Vendors shall provide evidence supporting their claimed performance from operational (not laboratory-only) deployments.

Scope of detection rate: Detection rate applies to faults that produce electrically observable signatures - i.e., faults that modulate motor current, voltage, or power in a manner detectable by spectral or statistical analysis. Purely mechanical failures with no torque coupling or electrical manifestation (e.g., external structural corrosion, non-load-bearing component wear) are outside the scope of ESA and shall not be counted against vendor detection-rate performance.

Metric	Target	Evidence / Qualification
Fault Detection Rate	>85% of developing electrically observable faults across all four categories	Measured over a minimum 12-month evaluation period on production assets. Vendor to provide methodology for calculating detection rate, including definition of 'developing fault' and how missed faults are counted. The denominator shall include only faults with confirmed electrical or torque-modulation signatures.
False-Positive Rate (validated alerts)	<2% (target <1%)	Calculated as: validated alerts that did not correspond to a confirmed fault / total validated alerts issued. Vendor to report from operational deployments, not laboratory trials.
Detection Lead Time	≥4 weeks advance warning (typical)	Lead time varies by fault type and severity. 4 weeks is a reasonable expectation for gradually developing mechanical faults. Vendor to provide case studies with documented timelines.
Diagnostic Specificity	Identification of fault type and likely location	Generic "motor anomaly" alerts are not acceptable. Alerts shall classify the detected condition within the defined fault categories (Electrical Faults, Mechanical Faults, Transmission Faults, or Load / Process Faults) and identify the probable fault mechanism (e.g., "pulley misalignment", or "pump blockage") to enable targeted maintenance action.

Exclusion Criteria — Fault Detection

- ✖ Generic fault models only (e.g., 'motor defect' without fault-type identification).
- ✖ Electrical-only focus with no mechanical or process fault coverage.
- ✖ Vibration-analysis algorithms directly repurposed for ESA without ESA-specific validation.
- ✖ No documented field evidence of specific fault-type detection.
- ✖ Claims of >95% detection rate without published methodology or independent validation.

3. Analytics and Diagnostics Requirements

The analytics engine determines whether an ESA system delivers actionable insight or unusable noise. This section specifies outcome-based requirements that are agnostic to the specific algorithm family employed.

3.1 Why Adaptive Diagnostics Are Essential

ESA fault signatures manifest as gradual spectral-shape changes and slow amplitude drifts, not sudden threshold exceedances. A threshold set on an absolute current-harmonic amplitude will fire differently depending on load, supply voltage, ambient temperature, and process state — none of which are fault-related.

Effective diagnostics must therefore:

- Learn a per-asset baseline under normal operating conditions.
- Track deviations from that baseline over time (trend analysis).
- Distinguish load-driven variation from fault-driven spectral change.
- Adapt when operating conditions change (e.g., seasonal load shifts, process recipe changes).

These principles can be satisfied by multiple algorithm families: physics-based / model-based approaches (e.g., motor equivalent-circuit models), statistical trend methods, deterministic baseline-learning algorithms, and supervised or unsupervised machine-learning models. This specification does not prescribe a particular approach. It requires demonstrated adaptive behaviour and validated field outcomes.

3.2 Analytics Requirements

Requirement	Specification	Rationale
Analysis Approach	**[MANDATORY]** Adaptive diagnostics with per-asset baseline learning. Fixed-threshold-only systems are excluded.	Fixed thresholds cannot separate normal operating variation from developing faults in ESA. Any approach — model-based, statistical, or ML — is acceptable provided it demonstrates adaptive baseline tracking.
Operating-Condition Awareness	**[MANDATORY]** Diagnostics must account for load state, supply conditions, and VFD operating point.	A 3–5% increase in current harmonics may be expected under changing load conditions or speed variations (e.g., due to VFD frequency adjustments). The same increase under stable operating conditions may indicate a developing fault. Analytics must distinguish between these cases.
VFD Compatibility	**[MANDATORY]** Full diagnostic capability on VFD-controlled motors, with documented approach to harmonic separation.	VFD switching harmonics mask fault signatures. The vendor must be able to demonstrate proven success through customer references.

Requirement	Specification	Rationale
Field Validation Evidence	**[MANDATORY]** Vendor shall provide case studies from production deployments with confirmed fault events.	Laboratory or simulation-only validation does not prove robustness under real-world noises, load variation, and equipment diversity.
Baseline Learning Period	**[PREFERRED]** Automatic baseline establishment within 2–4 weeks of installation.	Rapid baselining reduces time-to-value. Vendor to state expected learning period and any manual configuration required.
Training Data Breadth	**[PREFERRED]** Analytics developed or validated on data from ≥1,000 diverse industrial assets.	Broader exposure improves generalisation across motor types, sizes, and applications. Vendor to describe dataset composition and scale.
Continuous Improvement	**[PREFERRED]** Analytics platform supports model updates informed by newly confirmed fault events.	Static models degrade as equipment populations and process conditions evolve.

Exclusion Criteria — Analytics

- ✖ Fixed-threshold-only systems marketed as ‘anomaly detection’.
- ✖ Systems with no field-validated approach to VFD harmonic separation.
- ✖ Models validated only in laboratory or simulation environments.
- ✖ Vibration-analysis algorithms applied directly to current/voltage data without ESA-specific adaptation.
- ✖ No mechanism for model update or refinement post-deployment

4. Service Delivery Requirements

Condition monitoring programmes fail more often from poor alert management than from poor sensing. This section specifies how ESA insights must be validated, contextualised, and delivered.

4.1 Human-in-the-Loop Validation

Mandatory requirement: AI- or algorithm-generated fault alerts shall be reviewed by qualified condition monitoring engineers before notification to the customer when required to reduce false positives. Fast-developing faults may be automated.

Rationale:

- Automated diagnostics can misinterpret process changes, operational transients, or data-quality artefacts as faults.
- Expert review catches edge cases and ambiguous signatures that algorithms handle poorly.
- Validated alerts carry higher trust with maintenance teams, reducing alarm fatigue.
- Validation enables contextualised recommendations specific to the fault, equipment, and site.

Advisory status and responsibility: All validated alerts and maintenance recommendations issued by the monitoring service are advisory in nature. They are intended to support, not replace, the asset owner's maintenance decision-making processes. Final decisions regarding maintenance actions, asset operation, and resource allocation remain solely with the asset owner. The monitoring service provider shall not be held liable for maintenance decisions made or not made by the asset owner on the basis of advisory alerts.

Requirement	Specification	Rationale
Expert Validation	**[MANDATORY]** All fault alerts reviewed by qualified engineers before customer notification.	Reduces false positives and ensures alerts are actionable. Target: <2% false-positive rate on validated alerts (based on industry-reported benchmarks for expert-reviewed ESA programmes).
Response Time	**[MANDATORY]** Critical alerts validated and communicated within 48 hours of detection.	Timely notification enables planned intervention. 48 hours balances urgency with thorough review.
Ongoing Support	**[PREFERRED]** Access to expert helpdesk for alert contextualisation and maintenance planning during office hours.	Maintenance teams benefit from support translating validated alerts into effective work orders.

4.2 Actionable Output Requirements

Requirement	Specification	Rationale
Alert Content	**[MANDATORY]** Alerts include: specific fault type, severity level, recommended	Vague alerts ('possible motor issue') are not actionable.

Requirement	Specification	Rationale
	actions (advisory), and supporting evidence (spectral data or trend plots).	Specific diagnosis enables targeted repair.
Fleet Prioritisation	**[MANDATORY]** Priority ranking across all monitored assets based on fault severity.	Large fleets generate many alerts; prioritisation directs limited maintenance resources to highest-impact issues.
Maintenance Recommendations	**[PREFERRED]** Alerts include specific maintenance actions (advisory), not just fault identification.	Transforms monitoring from sensor data into a maintenance decision-support tool.
Historical Context	**[PREFERRED]** Asset health timeline showing fault progression and prior maintenance events.	Trend context improves repair-vs-replacement decisions and maintenance scheduling.

Exclusion Criteria — Service Delivery

- ✖ Raw algorithm alerts sent directly to customers without expert review if this leads to false alerts.
- ✖ Self-service-only platforms where customers must interpret ESA spectra unaided.
- ✖ Alerts without recommended actions or fault-type identification.

5. Monitoring Mode Requirements

The monitoring mode — continuous versus periodic — determines which faults can be detected and how much advance warning is achievable.

Requirement	Specification	Rationale
Monitoring Type	**[MANDATORY]** Permanent, online monitoring for critical and high-criticality assets. Periodic or portable monitoring may be acceptable for non-critical assets only (to be defined per site).	Industry experience indicates continuous monitoring catches significantly more developing faults than quarterly or monthly inspections.
Data Continuity	**[MANDATORY]** Local edge buffering to prevent data loss during network outages.	Connectivity gaps create diagnostic blind spots. Local buffering must retain at least 72 hours of acquisition data.
Operating-Condition Coverage	**[PREFERRED]** Monitoring during startup, steady-state, and shutdown (where applicable).	Startup current analysis is particularly valuable for rotor-bar defect detection. Faults may manifest only under specific operating conditions.
Load Variation Handling	**[MANDATORY]** The system must distinguish normal load variation from fault signatures.	Process-driven load changes modulate current spectra identically to some fault types. Analytics must separate these.

Exclusion Criteria — Monitoring Mode

- ✖ Portable-only or periodic-only monitoring for critical assets.
- ✖ Systems requiring manual data collection via walk-around.
- ✖ No local data buffering capability.

6. Deployment and Installation Requirements

ESA's primary physical advantage is non-intrusive installation at the Motor Control Cabinet (MCC), without requiring access to the asset itself. Deployment requirements should maximize this advantage.

Requirement	Specification	Rationale
Installation Location	**[MANDATORY]** All sensors and DAQ at the MCC. No on-asset sensors required.	MCC installation enables monitoring of submerged, enclosed, ATEX-zone, and otherwise inaccessible assets.
Installation Time	**[PREFERRED]** <60 minutes per asset typical; <90 minutes maximum.	Fast installation minimises disruption and enables rapid fleet rollout.
Environmental Rating	**[MANDATORY]** IP20 minimum for MCC-mounted equipment.	MCC enclosure provides primary environmental protection. IP20 protects against solid objects >12 mm.
Power Supply	**[MANDATORY]** Mains-powered or external DC supply. No battery-only sensors.	Battery-only devices require replacement cycles and risk data gaps when depleted.

Exclusion Criteria — Deployment

- ✖ Systems requiring on-asset sensor installation (negates ESA accessibility advantage).
- ✖ Battery-only sensors without mains-power option.
- ✖ Limited motor-type compatibility (e.g., DOL only).

7. Connectivity and Integration Requirements

Requirement	Specification	Rationale
Connectivity	**[MANDATORY]** Ethernet and at least one of: 4G/LTE cellular, Wi-Fi. Edge gateway for site aggregation.	Flexible connectivity addresses varied site IT infrastructure. Cellular enables monitoring at remote or unmanned sites.
Platform Architecture	**[MANDATORY]** Cloud-hosted SaaS platform with web-based dashboards.	Cloud delivery enables centralised visibility, automatic updates, and reduced local IT burden.
CMMS Integration	**[MANDATORY]** REST API for integration with major CMMS/EAM systems (SAP PM, IBM Maximo, etc.).	Closes the loop between detection and maintenance action via automated work-order creation.
Mobile Access	**[MANDATORY]**Mobile-responsive dashboards and/or native apps with push notifications.	Field teams need mobile access at point of work.

Exclusion Criteria — Connectivity

- ✖ No cloud-to-cloud API (requires manual data export)
- ✖ No CMMS integration capability
- ✖ No mobile access

8. Scalability Requirements

Requirement	Specification	Rationale
Fleet Capacity	**[MANDATORY]** Platform supports ≥1,000 monitored assets per account, across multiple sites.	Large industrial operations have hundreds to thousands of motors. System must scale without performance degradation.
Multi-Site Visibility	**[MANDATORY]** Centralised dashboard with drill-down by region, site, area, and asset.	Distributed operations need unified cross-site visibility.
Fleet Prioritisation	**[MANDATORY]** Automated alert prioritisation across entire fleet by fault severity.	Prevents important issues from being buried in alert volume.
Benchmarking	**[PREFERRED]** Cross-site and cross-industry benchmarking capabilities.	Identifies underperforming sites and supports best-practice sharing.
Deployment Speed	**[PREFERRED]** Vendor to state achievable rollout rate (assets per week), supported by reference evidence.	Slow deployment delays time-to-value. Vendor claims must be backed by documented project evidence.
Reference Deployments	**[MANDATORY]** Vendor demonstrates existing deployments at comparable scale (≥1,000 assets) for a single client.	Pilot-scale success does not guarantee enterprise scalability.

Exclusion Criteria — Scalability

- ✖ No enterprise-scale reference deployments
- ✖ No fleet-level alert prioritisation

9. Security and Compliance Requirements

Requirement	Specification	Rationale
Data Encryption	**[MANDATORY]** TLS 1.2+ in transit; AES-256 at rest.	Protects operational data from interception and unauthorised access.
Security Certification/Attestation	**[MANDATORY]** Valid ISO 27001 and/or SOC 2 Type II, where the ESA-based condition monitoring services must be included in the scope Where ISO 27001 is submitted, it must be accompanied by the Statement of Applicability document. Where SOC 2 is submitted, the report must cover at minimum the Security, Confidentiality and Availability trust services criteria. Partial SOC 2 reports covering fewer principles are insufficient.	Third-party certification/attestation of in scope services validates security practices against international standards. Requiring the Statement of Applicability prevents narrow ISO 27001 submissions that exclude too many security controls. Requiring Security, Confidentiality + Availability prevents narrow SOC 2 submissions that omit protection of customer data or uptime assurance.

Requirement	Specification	Rationale
Penetration Testing	**[PREFERRED]** Annual independent penetration testing of IoT and cloud infrastructure.	Identifies vulnerabilities before exploitation.
Access Control	**[MANDATORY]** Role-based access control (RBAC) with audit logging.	Controls data visibility; audit trails support compliance.
Authentication	**[MANDATORY]** SSO integration (SAML 2.0 / OAuth 2.0) and MFA support.	Integrates with enterprise identity management and reduces credential-compromise risk.
Data Residency	**[PREFERRED]** Configurable data-residency options (regional cloud hosting).	Addresses regulatory and policy requirements for data location (e.g., GDPR).

Exclusion Criteria — Security

- ✖ No recognised or invalid security certification and/or attestation
- ✖ The ESA-based condition monitoring service not included in the scope of the submitted security certification and/or attestation
- ✖ No encryption in transit or at rest
- ✖ No SSO or MFA capability
- ✖ ISO 27001 certification not accompanied by Statement of Applicability document
- ✖ SOC 2 report that does not include Security, Confidentiality and Availability trust principles

10. Hardware Certification Requirements

Certification	Requirement	Rationale
CE Marking	**[MANDATORY]** All hardware carries CE marking.	EU conformity for health, safety, and environmental standards.
UL / IEC Listing	**[MANDATORY]** UL listing or equivalent IEC 61010 compliance for electrical safety.	Independent verification of electrical safety in industrial environments.
EMC Compliance	**[MANDATORY]** Compliance th EN 61326-1 or equivalent for industrial EMC.	Ensures hardware operates reliably in electrically noisy industrial settings without introducing interference.

11. Vendor Qualification Requirements

Requirement	Specification	Rationale
ESA Experience	**[MANDATORY]** Minimum 3 years delivering ESA-based condition monitoring in production environments.	ESA requires deep domain expertise in signal processing, rotating-equipment failure modes, and analytics development.
Installed Base	**[PREFERRED]** ≥1,000 assets monitored globally.	Scale indicates market validation and provides data diversity for analytics improvement.

Requirement	Specification	Rationale
Industry References	**[MANDATORY]** Minimum 3 referenceable customers in comparable industries.	Peer references validate real-world capability.
Training & Support	**[MANDATORY]** Training programme, user documentation, and ongoing technical support.	Ensures customer teams can operate the system effectively.
Financial Stability	**[PREFERRED]** Evidence of financial viability (revenue trajectory, funding, or strategic backing).	Long-term monitoring programmes require vendor continuity.
OEM Partnerships	**[PREFERRED]** Partnerships with motor/drive OEMs.	Indicates technology validation and deployment synergies.

Pre-Evaluation Disqualification Checklist

Before scoring, evaluators shall screen all proposals against the following mandatory-exclusion criteria. A proposal that fails any item below shall be disqualified from further evaluation. This checklist exists to prevent scoring effort on fundamentally non-compliant submissions.

#	Disqualification Criterion	Rationale
DQ-1	System provides RMS-only or averaged metrics without waveform-level spectral analysis capability.	RMS values discard the spectral information essential for ESA fault detection.
DQ-2	System uses energy-disaggregation or NILM approaches that infer individual machine state from aggregate power consumption.	NILM cannot provide the per-asset spectral resolution required for fault-type identification.
DQ-3	No voltage measurement capability where full ESA is required by this specification.	Current-only systems cannot perform ESA. See Section 1 and Appendix B.3.
DQ-4	No field-validated case studies provided. All evidence is from laboratory, simulation, or marketing materials only.	Field validation is a mandatory requirement (Section 3.2). Unvalidated systems present unacceptable risk.
DQ-5	No documented VFD diagnostic capability or evidence of operation on VFD-controlled motors.	VFD compatibility is mandatory (Section 3.2). Systems untested on VFDs cannot be deployed in typical industrial environments.
DQ-6	Fixed-threshold-only analytics with no adaptive or baseline-learning mechanism.	Fixed thresholds are explicitly excluded (Section 3.2).
DQ-7	No human-in-the-loop expert validation of alerts prior to customer notification.	Expert validation is mandatory (Section 4.1).

Procedure: Evaluators shall complete this checklist for each proposal before proceeding to scoring. Any proposal failing one or more criteria shall be returned to the vendor with a written explanation of the disqualification ground(s). The vendor may resubmit with corrected evidence within the tender response window if permitted by procurement rules.

Evidence Hierarchy for Evaluation Scoring

Evaluators shall weight vendor evidence according to the hierarchy below. Higher-tier evidence carries greater weight in scoring. Marketing claims, brochure statements, or unsubstantiated assertions without supporting evidence shall score zero.

Tier	Evidence Type	Description	Scoring Guidance
1	Published field case studies	Documented deployments describing fault type, detection timeline, asset context, and outcome. Published in industry journals, conference proceedings, or detailed customer-authorised reports.	Highest weight. Directly demonstrates real-world capability. Multiple case studies across fault types strengthen the score.
2	Reference customer calls	Evaluator-led calls with named reference customers who confirm fault detection, service quality, and operational experience at comparable scale.	High weight. Provides independent peer validation. At least 2 reference calls required for top-tier scores.
3	Independent third-party validation	Testing, audit, or benchmarking performed by an independent body (university, certification lab, industry consortium) with published or auditable results.	Moderate weight. Valuable when field case studies are limited. Must be genuinely independent (not vendor-funded without disclosure).
4	Internal tests and simulations	Vendor-conducted laboratory tests, simulation results, or bench demonstrations without independent oversight or field confirmation.	Lowest weight. Useful for demonstrating technical approach but insufficient alone for scores above 3.

Zero-score rule: Marketing claims, product brochures, and self-declared capabilities without any supporting evidence from the tiers above shall receive a score of 1 (does not meet). Evaluators are not required to infer capability from marketing materials.

Appendix A: Evaluation Scoring Matrix

Use this matrix to score vendor proposals. Weights reflect critical success factors for ESA-based condition monitoring. Adjust weights to reflect organisation-specific priorities. Apply the Evidence Hierarchy (above) when assigning scores.

Evaluation Category	Weight	Score (1–5)	Weighted Score	Evidence Required
1. Signal Acquisition Quality	10%			ADC specs (ENOB), sampling rate, channel configuration, anti-aliasing design
2. Fault Detection Coverage	15%			Fault-type list, detection-rate methodology, field case studies
3. Analytics / Diagnostics Capability	20%			Approach description, VFD handling, baseline-learning evidence, field validation
4. Service Delivery (Human-in-Loop)	20%			Validation workflow, staff qualifications, false-positive rate evidence
5. Monitoring Mode (Continuous)	10%			Architecture description, edge buffering, startup-analysis capability
6. Scalability & Deployment	10%			Reference deployments at scale, rollout rate evidence
7. Connectivity & Integration	5%			API documentation, CMMS integration references
8. Security & Compliance	5%			Certifications, pen-test reports, encryption specifications
9. Vendor Qualification	5%			References, installed base, financial stability evidence
TOTAL	100%			

Scoring Guide

- **5 — Exceeds requirement:** Clear evidence of capability beyond specification. Tier 1 or 2 evidence required.

- **4 — Fully meets requirement:** Complete compliance with supporting evidence. Tier 1 or 2 evidence required.
- **3 — Substantially meets:** Minor gaps, acceptable with documented mitigation plan. Tier 2 or 3 evidence minimum.
- **2 — Partially meets:** Significant gaps requiring workarounds or development. Tier 3 or 4 evidence only.
- **1 — Does not meet:** Fails to address requirement, or no evidence beyond marketing claims.

Scoring principle: Scores shall be based on documented evidence (case studies, datasheets, reference calls), not on vendor claims alone. A score of 4 or 5 requires Tier 1 or Tier 2 evidence as defined in the Evidence Hierarchy.

Appendix B: Signal Quality Technical Reference

This appendix provides technical background on signal-quality requirements for engineering evaluators. Values stated here are consistent with, and provide justification for, the main-body requirements in Section 1.

B.1 Nyquist Sampling and Frequency Resolution

Two distinct signal-processing constraints govern acquisition design: bandwidth (Nyquist) and spectral resolution.

Bandwidth (Nyquist Theorem)

The Nyquist–Shannon theorem requires a sampling rate of at least 2× the highest frequency of interest. For ESA fault detection:

- Primary fault frequencies (rotor-bar sidebands, eccentricity, stator faults): 0–2.5 kHz. Minimum sampling: 5 kHz.
- Bearing-related load-modulation harmonics: typically 1–5 kHz, occasionally to 10 kHz. Minimum sampling: 10–20 kHz.
- High-order bearing-related harmonics and modulation effects: may extend to frequencies observable up to 20 kHz in current and power signatures. Minimum sampling: 40 kHz.

A 16 kHz sampling rate captures frequencies up to 8 kHz (with anti-aliasing), covering the primary motor-fault band and lower-order bearing-related harmonics. 20 kHz extends coverage to 10 kHz, encompassing the vast majority of ESA-detectable fault signatures. 40 kHz extends further to high-order modulation effects where electrically observable. The specification adopts a three-tier structure (16/20/40 kHz) to reflect that many proven ESA systems operate effectively at 16 kHz while higher rates provide progressively greater margin.

Spectral Resolution

Spectral (frequency) resolution is determined by the acquisition window duration, not the sampling rate:

$$\Delta f = 1 / T$$

where T is the acquisition window in seconds.

Window Duration (T)	Resolution (Δf)	Practical Significance
10 s	0.1 Hz	Resolves rotor-bar sidebands (2sf) for most 4-pole 50 Hz motors at typical slip values (≥1%).
15 s	0.067 Hz	Improves discrimination of closely spaced sidebands at low slip, increasing sensitivity to early-stage faults at low load conditions, without significantly increasing acquisition time.
60 s	0.017 Hz	Resolves close sidebands for slow-speed machines and low-slip conditions. Useful for 2-pole motors.
100 s	0.01 Hz	Required for 0.01 Hz resolution. Necessary only for very-low-slip or very-slow-speed applications.

Key point: Sampling rate and spectral resolution are independent parameters. High sampling rate provides bandwidth; long acquisition windows provide resolution. Both must be specified.

B.2 ADC Resolution: Nominal vs. Effective Bits

ADC manufacturers specify nominal bit depth, but real-world performance is lower due to thermal noise, integral and differential non-linearity, and clock jitter. The effective number of bits (ENOB) captures actual performance:

$$SQNR = 6.02 \times ENOB + 1.76 \text{ dB}$$

Nominal Bits	Theoretical SQNR	Typical ENOB (Industrial)	Effective SQNR	Practical Implication
12-bit	74 dB	10–11 bits	62–68 dB	Detects moderate-to-severe faults. May miss early-stage anomalies.
16-bit	98 dB	13–15 bits	80–92 dB	Suitable for most industrial ESA. Good early-fault sensitivity.
24-bit	146 dB	18–21 bits	110–128 dB	Highest sensitivity. Resolves very subtle fault signatures in the presence of dominant 50/60 Hz fundamental.

Specification requirement: Vendors shall state ENOB (or measured SINAD) at the actual acquisition sampling rate, not only nominal ADC bit depth.

B.3 ESA vs. MCSA: Why Voltage Measurement Matters

Motor Current Signature Analysis (MCSA) measures current only. Electrical Signature Analysis (ESA) measures both current and voltage. The practical differences:

Capability	MCSA (Current Only)	ESA (Current + Voltage)
Rotor-bar / eccentricity detection	Yes	Yes
Bearing fault detection (via load-modulation effects in current)	Yes (limited sensitivity)	Yes (improved via power analysis)
Supply voltage quality (unbalance, harmonics, sags)	No	Yes
Power factor and efficiency analysis	No	Yes
Real/reactive power calculation	No	Yes
VFD output-stage diagnostics	Limited	Yes (voltage waveform analysis)

Capability	MCSA (Current Only)	ESA (Current + Voltage)
Stator winding impedance asymmetry	Indirect only	Yes (via voltage/current comparison)

For comprehensive condition monitoring, full ESA (current + voltage) is required. Where only mechanical fault detection is needed and supply-quality monitoring is handled by other systems, current-only MCSA may be acceptable—but this limitation must be explicitly acknowledged in the vendor proposal.

Appendix C: Scope Boundaries of ESA

This appendix defines the known limitations of Electrical Signature Analysis to set realistic expectations and prevent vendor over-claiming. It is included to strengthen the credibility of this specification and to ensure that evaluation criteria are applied fairly.

ESA is a proven and valuable condition monitoring technique, but it is not omniscient. The following conditions reduce or eliminate ESA diagnostic capability:

C.1 Faults with No Electrical or Torque Coupling

ESA detects faults through their effect on motor current, voltage, or power signatures. Faults that produce no measurable change in these electrical quantities are outside ESA's scope. Examples:

- External structural corrosion or erosion that does not affect rotating components or load
- Seal leaks or gasket failures that do not change the driven load
- Non-load-bearing component degradation (mounting bolts, covers, guards) until it produces secondary vibration or load change
- Lubrication degradation in very early stages before it creates detectable friction or load-modulation effects

These fault types require complementary monitoring methods (visual inspection, vibration, ultrasound, thermography, or oil analysis).

C.2 Very Low-Load or Unloaded Operating Conditions

ESA relies on load-torque modulation to transfer mechanical fault signatures into the current spectrum. When a motor operates at very low load (typically <15–20% of rated load), the coupling between mechanical faults and current modulation weakens significantly. Under these conditions:

- Bearing-related signatures may fall below the noise floor
- Transmission fault signatures (belt, gearbox) become undetectable if no meaningful torque is transmitted
- Rotor-bar fault sidebands compress toward the supply frequency as slip approaches zero

Vendors should state the minimum load threshold at which their diagnostics are effective, and monitoring strategies should account for variable-load duty cycles.

C.3 Highly Distorted or Unavailable Electrical Signals

ESA diagnostic quality depends on access to clean, representative electrical signals. Diagnostic capability may be reduced or unavailable when:

- Current or voltage signals are severely distorted by upstream power-quality issues unrelated to the monitored asset
- VFD output waveforms are so heavily modulated that fault-frequency sidebands cannot be separated from switching harmonics (despite vendor mitigation methods)
- Only partial measurement access is available (e.g., single-phase measurement on a 3-phase system, or current-only where voltage is required)

- Signal conditioning or CT/VT instrumentation introduces significant noise or bandwidth limitations

These limitations do not indicate a failure of the ESA approach. They define the boundaries within which ESA operates reliably, and they should inform asset-selection decisions during deployment planning.

C.4 Complementary Monitoring

For assets or fault types outside ESA's reliable detection scope, complementary monitoring methods should be considered. ESA is most effective when deployed as part of a broader reliability strategy that may include vibration analysis, thermography, ultrasound, oil analysis, and visual inspection as appropriate to the asset and operating context.

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